Formulas of the InferMesh Gateway

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Abstract—InferMesh is an OpenAI-compatible gateway that sends each request to an LLM worker likely to hold the prompt prefix in KV cache, and that keeps serving when a worker fails or rate-limits. Nothing in the live path is a new theorem: each identity is a published special case (SGLang cache-aware routing, Azar–Mitzenmacher power of two choices, the SRE 10% retry budget, Envoy linear ejection time, RFC 9110 Retry-After) or is marked assumed. This note records the identities the code computes, the defaults, and the three places the gateway differs from the cited source (miss → P2C, ETIF mix $\alpha = 0.6$, DistServe C_i in the replay only).

Index Terms—LLM serving, prefix cache, power of two choices, retry budget, outlier ejection

I. Introduction

A transformer engine reuses a prompt prefix only if the next request lands on the worker that still holds those KV blocks. Independent uniform assignment wastes that reuse; sticky assignment pins load. Failures and 429s add a second problem: unbounded retries turn a short outage into a metastable collapse [8].

InferMesh sits between OpenAI-format clients and OpenAI-compatible backends. On each request it (i) matches the prompt in a character radix trie, (ii) picks a worker, (iii) admits at most one retry from a credit ledger, (iv) ejects a worker that posts consecutive 5xx. The predicates and constants are those of SGLang's cache-aware router [9], Azar et al. [1] with Mitzenmacher's supermarket model [2], Google SRE Chapter 21 [4], and Envoy outlier detection [5]. Replay service times follow the form of DistServe Appendix A [7] with unpublished constants.

This is a specification of the running code, not a claim of a new algorithm. A line marked assumed is used and has no published derivation. Simulator-only identities are confined to Section XI. Defaults below are the shipped values.

a) Notation.: Pool W for a model: the breaker admits the worker and waiters $< Q$. Load $L(w)$ is requests in flight. Prompt string p. Match depth d. Attempts $A = 2$.

II. Prefix

$$p = \text{join}(\mathsf{role} : \mathsf{content})$$

over messages, newline-separated (extract_prefix). Completions use the raw prompt string. Embeddings skip the trie.

III. Token reserve (ETIF)

Heuristic, not a tokenizer. Let c be the sum of content lengths in messages (roles are not counted).

$$\hat{n} = \max\big(1, \lfloor c/4 \rfloor\big) + \texttt{max_tokens}$$

(or 256 if the client omits max_tokens). Set etif_count_output_tokens to false to drop the second term. Reserved until the response ends. c is not $|p|$: the trie string includes role:.

IV. Selection

Problem. Send p to a worker that already holds a long prefix of p, unless that pins load.

SGLang cache-aware (sgl-model-gateway) uses CLI defaults cache-threshold 0.3, balance-abs-threshold 64, balance-rel-threshold 1.5; load is in-flight count. Imbalanced iff both

$$\max L - \min L > 64 \quad \text{and} \quad \max L > 1.5 \min L.$$

Otherwise a hit iff there exist live tenants of the longest match with

$$d \geq d_{\min} \quad \text{and} \quad d > 0.3|p|,$$

where $d_{\min}$ is kv_match_min_chars. Then pick $\arg\min L$ among those tenants.

Here: the same predicates. On miss or imbalance, power-of-two choices, not SGLang's min-load (random tie-break). Match rate in SGLang is $d/|p|$ in characters; same here. Insert of p on dispatch (insert_on_dispatch=true), as SGLang inserts at selection.

V. Power of two choices

Theorem 1 (Azar–Broder–Karlin–Upfal [1]): n balls, n bins, each ball into the least loaded of $d \geq 2$ uniform bins: max load $(\log\log n)/(\log d) + O(1)$ w.h.p. For $d = 1$ it is $\Theta(\log n / \log\log n)$.

Theorem 2 (Supermarket [2]): Poisson arrivals rate λn, $\lambda < 1$, exponential service; $d = 2$ cuts expected wait exponentially versus $d = 1$.

Here: $d = 2$. Sample without replacement, weights = worker weight. Compare scores, not raw queue length. The theorems apply only while the score is a live queue; they do not cover a stale or mixed score.

VI. Score

Let $T(w)$ be tokens in flight, $U(w) \in [0,1]$ engine KV util, $q(w)$ engine wait depth. Write $e = \min(T/S, 1)$ and $q' = \min(q/q_0, 1)$. Then

$$s(w) = \begin{cases} e & \text{openai or stale } U, \\ \alpha e + \beta q' + (1-\alpha-\beta)U & \text{otherwise.} \end{cases}$$

Defaults: $\alpha = 0.6$, $\beta = 0$, $q_0 = 8$, $S = 20 \cdot 1024$ (or etif_scale). Stale means older than 3 poll intervals. $\alpha = 0.6$ is assumed (no published mix). With $\beta = 0$ the vLLM term is $0.4U$.

VII. Trie

Character radix (Morrison PATRICIA [3]): one child per first character, split on LCP. Tenants stamped on every node of an insert. Match may stop inside an edge.

TTL 300 s per attribution. Cap 2^{26} characters per worker (SGLang max_tree_size). Over cap: drop LRU leaves until 0.9 of the cap (assumed headroom; SGLang uses a periodic node LRU). LCP length: binary search on slices, $O(\log n)$ Python steps, $O(n)$ bytes compared.

VIII. Retry budget

SRE [4]: retry while the retry ratio stays below 10 percent; amplification at most about 1.1 versus about 3 with a per-request cap of three. Envoy RetryBudget: default 20 percent of active+pending, floor min_retry_concurrency=3.

Here: credit ledger, mode static,

$$c \leftarrow \min(c_{\max},\, c + 0.10),$$
$$\rho = \rho_0(1 - \alpha_\rho h), \quad \rho_0 = 1,\ \alpha_\rho = 0.$$

Admit a retry iff $c \geq \rho$; then $c \leftarrow c - \rho$. With $\alpha_\rho = 0$, $\rho = 1$: one retry per ten requests in the mean, burst $c_0 = 3$, cap $c_{\max} = 10$. $h = d_w/|p|$ for the fallback worker; unused while $\alpha_\rho = 0$.

Mode headroom (assumed, off): deposit $\max\big(0, \min(R/(\rho_0\lambda) - 1, 1)\big)$ once λ is known over a window of at least 1 s. Local reroutes (open breaker, full queue) skip the ledger. Hard cap $A = 2$ attempts, distinct workers. Retry only before the first byte, on 429, 5xx, or transport. Timeouts are not retried.

IX. Circuit breaker

Envoy outlier [5]: consecutive_5xx=5. Real ejection time equals base_ejection_time times consecutive ejections, capped by max_ejection_time. Defaults 30 s and 300 s.

$$T(k) = \min(T_0 k,\, T_{\max}), \qquad T_0 = 30,\ T_{\max} = 300.$$

First open $k = 1 \Rightarrow T = 30$ s. Success sets $k = 0$. Explicit Retry-After cool-down does not increment k.

States: CLOSED → OPEN after 5 consecutive 5xx (not 429) → HALF_OPEN after T, one probe. 429 opens for the header delay, unmultiplied.

Not taken from Envoy: max_ejection_percent (default 10 percent); the interval that decrements the multiplier while healthy (here a success zeros k).

X. Cooldown and queue

RFC 9110 §10.2.3 [6]: Retry-After is an HTTP-date or delay-seconds. Invalid ⇒ ignore. Else the largest vendor reset (x-ratelimit-reset-*, Anthropic RFC 3339 / epoch). Cap 300 s. Never ∞.

Semaphore $C = 20$ in-flight per worker. Waiters $\leq Q = 40$, wait ≤ 30 s, else 503 with Retry-After. Routable while waiters $< Q$.

XI. Replay benches (not the gateway)

Prefill time, DistServe [7] Appendix A form $T = C_1(\cdots) + C_2(\cdots) + C_3$, constants fitted per model/GPU and unpublished:

$$t_{\text{prefill}}(u) = c_1 u + c_2 u^2 + c_3$$

with $c_1 = 4 \cdot 10^{-5}$, $c_2 = 3 \cdot 10^{-9}$, $c_3 = 0.02$, ITL 0.02 s assumed. Claim policy order, not seconds.

SLO in the replay metrics: TTFT 0.25 s, TPOT 0.10 s (DistServe Table 1, OPT-13B chatbot). docs/benchmarks.md reports a 2 s TTFT cut on the long Mooncake prompts. Retry sim: Bronson et al. [8] §2.1, capacity 300 QPS, offered 280 QPS.

XII. Conclusion

The live gateway is a product of published special cases. Cache hit and imbalance are SGLang's CLI predicates; $d = 2$ sampling is Azar–Mitzenmacher; the retry ledger is SRE's 10 percent ratio with $\rho = 1$; ejection time is Envoy's $T = T_0 k$ with $T_0 = 30$ s and cap 300 s; Retry-After is RFC 9110.

Three extras remain and are labelled as such: P2C on miss (SGLang uses min-load), the ETIF mix $\alpha = 0.6$, and DistServe C_i in the replay only. α_ρ and headroom mode stay off. Those constants are not fitted here.

References

[1] Y. Azar, A. Z. Broder, A. R. Karlin, and E. Upfal, "Balanced allocations," SIAM J. Comput., vol. 29, no. 1, pp. 180–200, 1999.

[2] M. Mitzenmacher, "The power of two choices in randomized load balancing," IEEE Trans. Parallel Distrib. Syst., vol. 12, no. 10, pp. 1094–1104, 2001.

[3] D. R. Morrison, "PATRICIA—practical algorithm to retrieve information coded in alphanumeric," J. ACM, vol. 15, no. 4, pp. 514–534, 1968.

[4] B. Beyer, C. Jones, J. Petoff, and N. R. Murphy, Eds., Site Reliability Engineering. Sebastopol, CA: O'Reilly, 2016, ch. 21.

[5] Envoy Proxy, "Outlier detection," https://www.envoyproxy.io/docs/envoy/latest/intro/arch_overview/upstream/outlier.

[6] R. Fielding, M. Nottingham, and J. Reschke, Eds., "HTTP semantics," RFC 9110, Jun. 2022, §10.2.3.

[7] Y. Zhong et al., "DistServe: Disaggregating prefill and decoding for goodput-optimized large language model serving," in Proc. OSDI, 2024, arXiv:2401.09670.

[8] N. Bronson, A. Aghayev, A. Charapko, and T. Zhu, "Metastable failures in distributed systems," in Proc. HotOS, 2021.

[9] SGLang Project, "Cache-aware router," sgl-model-gateway, CLI defaults cache-threshold 0.3, balance-abs-threshold 64, balance-rel-threshold 1.5.